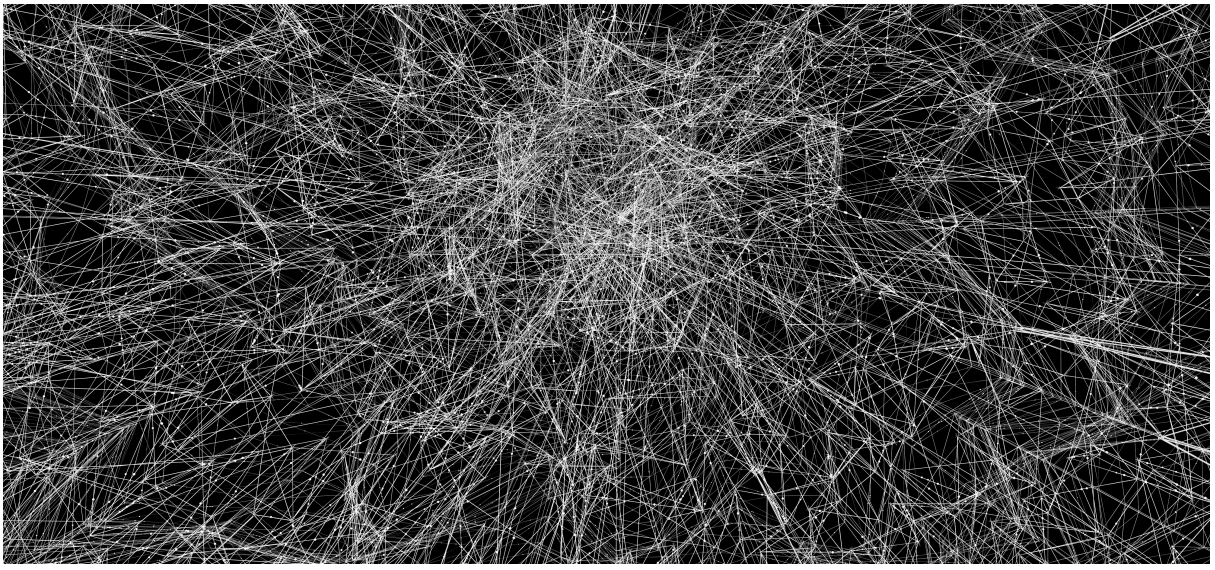


sVAD: Spiking Voice Activity Detection Reconstruction

1. We notice some parameters were listed.

Parameter	Value
Frame duration	30 ms
Frame overlap	50%
Frame shift	15 ms
Encoded features	20
Recurrent hidden neurons	32
Output neurons	2
LIF decay constant **α**	0.5
Firing threshold **θ**	0.3
Surrogate gradient parameter **a**	4
Batch size	128
Training epochs	100
Optimizer	Adam
Initial learning rate	0.001
LR schedule	Multiply by 0.1 every 40 epochs

▼ **Notes:**

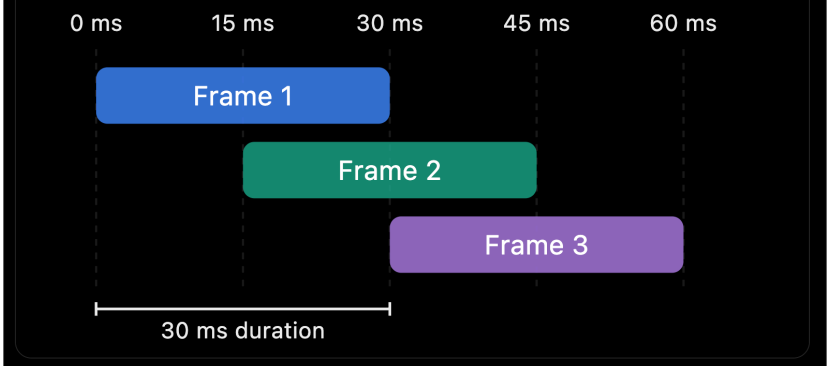
▼ Recurrent Diagonal vs Recurrent Normal Neural Network

In a recurrent diagonal network the feedback is only based on its own previous state. The weight matrix is thus a diagonal matrix.

▼ Frame Duration, frame overlap and frame shift

The paper uses:

- Frame duration = 30 ms
- Frame overlap = 50%
- Frame shift = 15 ms



Frame duration: How much audio is contained in each window. Here, each frame covers 30 ms.

Frame overlap: How much audio is shared between consecutive windows. A 50% overlap means consecutive 30 ms frames share 15 ms of audio.

Frame shift: How far we move forward before extracting the next frame.

consecutive 30 ms frames share 15 ms of audio. $\text{Frame shift} = 30(1 - 0.5) = 15\ \text{ms}$

Frame shift: How far we move forward before extracting the next frame.

$$\text{Frame shift} = 30(1 - 0.5) = 15\ \text{ms}$$

So we make a new speech/non-speech decision every 15 ms.

At a sample rate of 16 kHz, for example, one 30 ms frame contains 480 audio samples and the shift is 240 samples. The 16 kHz rate is an illustrative assumption, not a sampling rate explicitly specified in this paper's experimental setup.

▼ Encoded Features

Raw audio is just a sequence of amplitude values.

The auditory encoder transforms this waveform into more useful representations, such as learned responses to different frequency bands.

In sVAD, the encoder produces 20-dimensional features.

$$S[t] = [s_1, s_2, \ldots, s_{20}]$$

▼ Recurrent hidden neurons

The paper uses 32 recurrent spiking neurons in its classifier. These are neurons that maintain internal membrane states and have recurrent connections. These constitute the hidden layers.

▼ Output Neurons

2 linear readout neurons → [speech, not speech]. Then softmax function applied to create it into a probability. This then is used to train the module using cross entropy.

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

▼ Neural Network Constants (α, a)

$$U[t] = \alpha U[t-1] + I[t] - \vartheta O[t-1]$$

For the equation for Leaky Integrate and Fire NN

LIF Decay Constant (α): The decay constant α determines how much previous membrane potential is retained. A smaller α means the neuron forgets its previous voltage more quickly. A value closer to 1 means it retains it longer.

Surrogate Gradient Parameter (a): The spike function is a step function. Its true derivative is zero almost everywhere and undefined at the threshold, making ordinary backpropagation ineffective. Instead of using the actual derivative, it approximates it by:

$$\frac{\partial O}{\partial U} \approx \begin{cases} \frac{1}{a}, & |U - \vartheta| < \frac{a}{2} \\ 0, & \text{otherwise} \end{cases}$$

During the forward pass, the neuron still produces hard 0/1 spikes. The surrogate gradient is used only for calculating gradients during backpropagation.

▼ Everything else

Term	What it means	Paper's setting
Batch size	Number of training samples processed before each optimizer update. A sample may contain a sequence of frames.	128
Training epochs	Number of complete passes through the training dataset.	100
Optimizer	Algorithm that updates weights using computed gradients.	Adam
Initial learning rate	Initial scale of the optimizer's parameter updates.	0.001
LR schedule	Rule that changes the learning rate during training.	Reduce by a factor of 10 every 40 epochs

2. Meanwhile, a lot of parameters were publicly listed. For example:

▼ The number of SNN timesteps,

How long the spiking neural network operates before making a prediction. A neuron may receive insufficient input to fire at timestep 1, but accumulate enough membrane potential to fire at timestep 3. More timesteps can give the network more opportunity to accumulate evidence, but they also increase computation.

▼ Exact SincNet configuration,

SincNet is the first feature-extraction layer of the model. A conventional convolutional layer learns the individual values of a filter:

$$w = [w_0, w_1, \ldots, w_{L-1}]$$

A SincNet layer instead constructs band-pass filters using two learnable cutoff frequencies. The idealized band-pass filter is:

$$g(t) = 2f_H \operatorname{sinc}(2f_H t) - 2f_L \operatorname{sinc}(2f_L t)$$

This allows the network to learn which frequency regions are useful for detecting speech.

We must thus choose :

Parameter	Meaning
Number of filters	How many separate frequency bands are learned
Kernel length	How many waveform samples form each filter
Stride	How far the filter moves for each convolution output
Padding	How boundaries are handled
Cutoff initialization	Initial frequency ranges before training
Window function	How the finite-length filters are tapered
Sample rate	Frequency scale of the audio

What is missing: the exact SincNet filter count, length, stride, padding and initialization.

▼ sConv1D parameters,

Its purpose is to extract more complex auditory patterns and convert the representation into spikes.

$$I_c[t] = \sum_{j,k} W_{c,j,k} X_j[t-k] + b_c$$

For a normal `Conv1d` layer:

$$P = C_{\text{out}} C_{\text{in}} K + C_{\text{out}}$$
$$where,\ C_{\text{out}} - output\ channels, C_{\text{in}} - input\ channels, K - kernel\ size$$

What is missing: its kernel size, channel dimensions, stride, padding and whether it uses bias.

▼ Attention-layer structure,

Decision making about which auditory features to consider and which ones to not.

$$\hat{S} = Y \odot M$$

▼ Loss-weighting hyperparameter λ

▼ Stage 1 Findings

Objective: Reconstruct the architecture proposed in sVAD: A Robust, Low-Power, and Light-Weight Voice Activity Detection with Spiking Neural Networks (ICASSP 2024), using the reported parameter count and experimental results to constrain unspecified design choices.

1. Implemented architecture

Our initial PyTorch reconstruction contains:

- Auditory encoder: SincNet followed by spiking 1D convolution (sConv1D).
- Attention mechanism: Three spiking fully connected layers generating a feature mask.
- Classifier: Spiking recurrent neural network (sRNN) with 32 hidden neurons and 2 output neurons.
- Training: Cross-entropy classification loss combined with MSE attention loss.

2. Current configuration

Parameter	Value
Sample rate	16 kHz (assumed)
Frame duration / shift	30 ms / 15 ms
Encoded features	20
SincNet kernel size	101 (assumed)
sConv1D kernel size	3
Attention hidden widths	31, 31
Recurrent neurons	32
Recurrence type	Diagonal
LIF decay α	0.5
Firing threshold ϑ	0.3
Surrogate gradient parameter a	4
Loss weight λ	1.0 (assumed)

3. Parameter-count results

Our reconstruction

4313

Paper's reported count

~4,300

Component	Parameters
Encoder	1,260
Attention	2,283
Classifier	770
Total	4,313

The architecture search identified 1,214 configurations within ±100 parameters of the reported 4,300.

Several configurations contain exactly 4,300 parameters, demonstrating that parameter count alone cannot uniquely identify the original model.

4. Diagonal vs dense recurrence

Two recurrent structures were investigated:

Structure	Classifier parameters
Diagonal recurrence	770
Dense recurrence	1,762

The 992-parameter difference corresponds exactly to the difference between 32 diagonal recurrent weights and a full 32 × 32 recurrent matrix.

The paper's notation suggests diagonal recurrence, but does not establish it conclusively.

5. Implementation verification

Forward pass successful

Backpropagation successful

Optimizer update successful

Output tensor dimensions verified

The synthetic test produced:

$$L_{\text{CE}} = 0.62549, \qquad L_{\text{MSE}} = 0.42$$

$$L_{\text{total}} = 1.04549$$

These results confirm basic computational correctness, not detection accuracy.

6. Remaining uncertainties

Five important implementation details remain unresolved:

1. Number of internal SNN timesteps.
2. Exact SincNet configuration.
3. sConv1D kernel and channel configuration.
4. Attention hidden-layer dimensions and mask formulation.
5. Loss-weighting parameter λ.

Additionally, recurrent connectivity must be verified experimentally.

7. Current conclusion

Stage 1 successfully establishes a working, configurable sVAD reconstruction with a parameter count consistent with the published model.

However, the original architecture has not been uniquely identified, and no speech-detection performance has been measured.

Next stage: Prepare labelled clean/noisy audio, train shortlisted architectures, and compare their HTER against the paper's reported results:

Noise condition	Target HTER
Low noise	4.0%
Medium noise	11.9%
High noise	19.1%

The next objective is to move from parameter-count matching to experimental validation, using speech-detection accuracy and noise robustness to distinguish plausible architectures.

▼ Stage 2 Findings

Project: Reconstruction of *sVAD: A Robust, Low-Power, and Light-Weight Voice Activity Detection with Spiking Neural Networks* (ICASSP 2024)

Status: Completed

1. Objective

The objective of Stage 2 was to verify that our initial PyTorch reconstruction could process labelled audio, learn speech/non-speech distinctions, and perform correct frame-level classification.

We used synthetic tone-and-silence recordings before introducing real speech data. This allowed us to debug the architecture, training process, and evaluation pipeline under controlled conditions.

2. Implemented system

Our initial sVAD reconstruction contains three components:

- **Auditory encoder:** SincNet and spiking 1D convolution.
- **Attention network:** Three spiking fully connected layers.
- **Classifier:** Spiking recurrent neural network with 32 hidden neurons and 2 output neurons.

The model contains **4,313 trainable parameters**, consistent with the paper's reported approximately 4.3K parameters.

The encoder, attention, and classifier account for 1260, 2283, and 770 parameters, respectively.

Our audio preprocessing uses 16 kHz sampling, 30 ms frames, and a 15 ms frame shift. The 16 kHz sampling rate and several architectural details are reconstruction assumptions rather than verified specifications from the authors.

3. Initial training experiments

We examined the influence of class imbalance and the attention reconstruction loss.

The combined training objective was:

$$L = L_{\text{CE}} + \lambda L_{\text{MSE}}$$

Four initial experiments were performed.

$$Half\ Time\ Error\ Rate\ (HTER) = \frac{False\ Alarm\ Rate + Miss\ Rate}{2}$$

Experiment	Best HTER	Final HTER
Unweighted CE, λ=1	44.3%	50.0%
Unweighted CE, λ=0	43.8%	50.0%
Weighted CE, λ=1	45.1%	54.9%
Weighted CE, λ=0	**21.9%**	**21.9%**

The first two experiments demonstrated a classification collapse: the network eventually predicted almost every frame as speech.

Removing the MSE loss alone did not resolve this issue.

4. Class imbalance investigation

Inspection of the sampled training data revealed approximately 74.7% speech frames and 25.3% non-speech frames.

Consequently, a classifier predicting speech everywhere could achieve relatively high conventional accuracy despite an HTER of 50%.

We introduced class-weighted cross-entropy, initially using weights of approximately 1.98 for non-speech and 0.67 for speech.

Class weighting altered the classifier's behavior and reduced its tendency to predict speech for every frame. The strongest initial results were obtained using weighted CE with λ=0.

This provides evidence that the class distribution and training objective substantially influence the model's performance.

5. Extended 30-epoch experiment

Using weighted cross-entropy and λ=0, we extended training to 30 epochs.

The training loss decreased from 0.7014 at epoch 1 to 0.2402 at epoch 30.

The best validation performance occurred at epoch 20:

- **MR (Miss Rate): 9.5%**
- **FAR (False Alarm Rate): 0.0%**
- **HTER (Half-Total Error Rate): 4.8%**

The final epoch produced an HTER of 10.3%.

These results demonstrate meaningful learning on synthetic audio, although validation performance fluctuated. The validation set consisted of only three recordings, limiting statistical reliability.

The synthetic HTER cannot be directly compared with the paper's QUT-NOISE-TIMIT results.

6. Final verification experiment

We performed a fixed-batch overfitting test to verify that the network could learn a small, nearly balanced dataset.

The test contained four synthetic recordings, comprising 76 labelled frames:

- 40 speech frames
- 36 non-speech frames

After 120 optimizer updates:

Metric	Result
Initial training HTER	48.89%
Final training HTER	**1.39%**
Final cross-entropy loss	0.0313
True positives	40
False negatives	0
False positives	1
True negatives	35
Training accuracy	**98.68%**

The model correctly classified 75 of 76 frames.

Additional verification checks confirmed the implemented audio-framing and frame-label-alignment logic.

These results establish that the reconstruction can fit a fixed synthetic training batch.

7. Findings and limitations

Confirmed findings

1. The reconstruction successfully performs forward propagation, backpropagation, and optimizer updates.
2. The synthetic data pipeline and frame-label verification tests pass.
3. Class imbalance plausibly contributed to the earlier always-speech predictions.
4. Weighted cross-entropy with $\lambda=0$ produced our best synthetic validation results.
5. The model can nearly memorize a small, fixed, balanced dataset.

Remaining uncertainties

The precise SincNet configuration, spiking convolution parameters, attention structure, recurrent connectivity, internal SNN timesteps, and original loss weight remain unresolved.

Furthermore, all experiments in this stage used synthetic audio. We have not established performance on real speech, nor have we reproduced the original paper's evaluation protocol.

8. Conclusion and next steps

Stage 2 establishes a functional audio-processing and training pipeline for our 4,313-parameter sVAD reconstruction.

The network's successful fixed-batch learning experiment supports proceeding to genuine speech data, while the earlier classification collapse demonstrates the importance of dataset balance and careful training-objective selection.

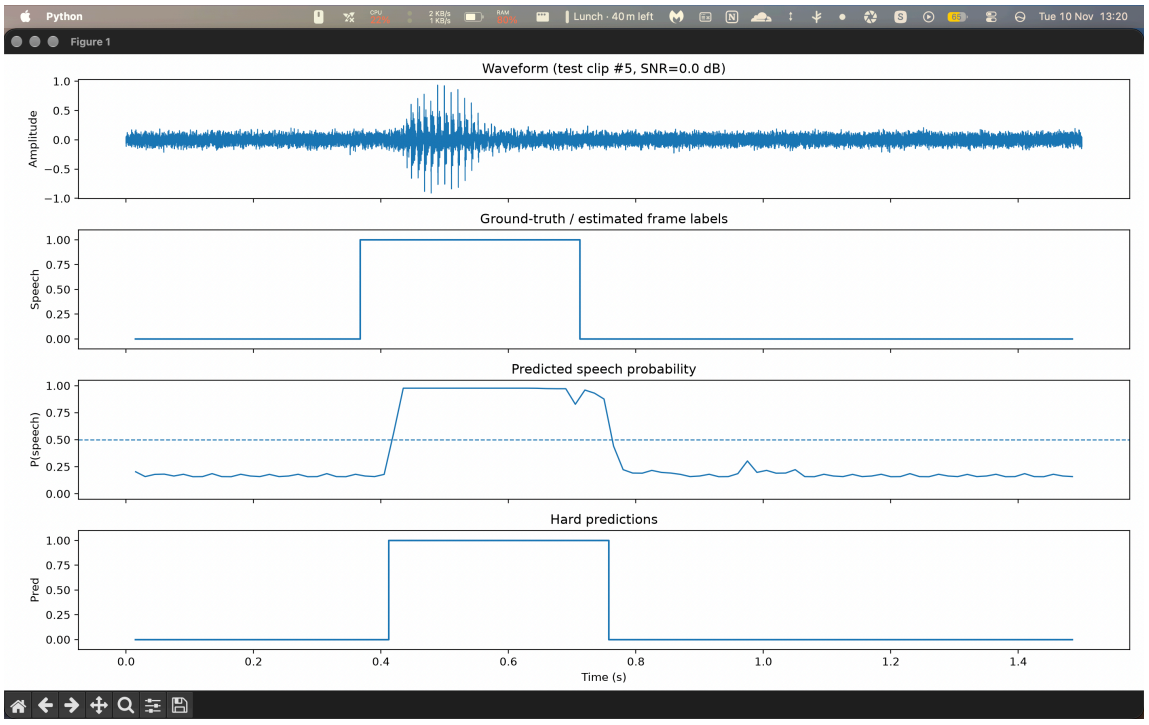
Stage 3 will focus on real speech recordings, clean/noisy audio pairs, noise robustness, fixed validation protocols, and evaluating architectural candidates against meaningful voice activity detection benchmarks.

The synthetic dataset will be retained as a debugging and regression-testing benchmark.

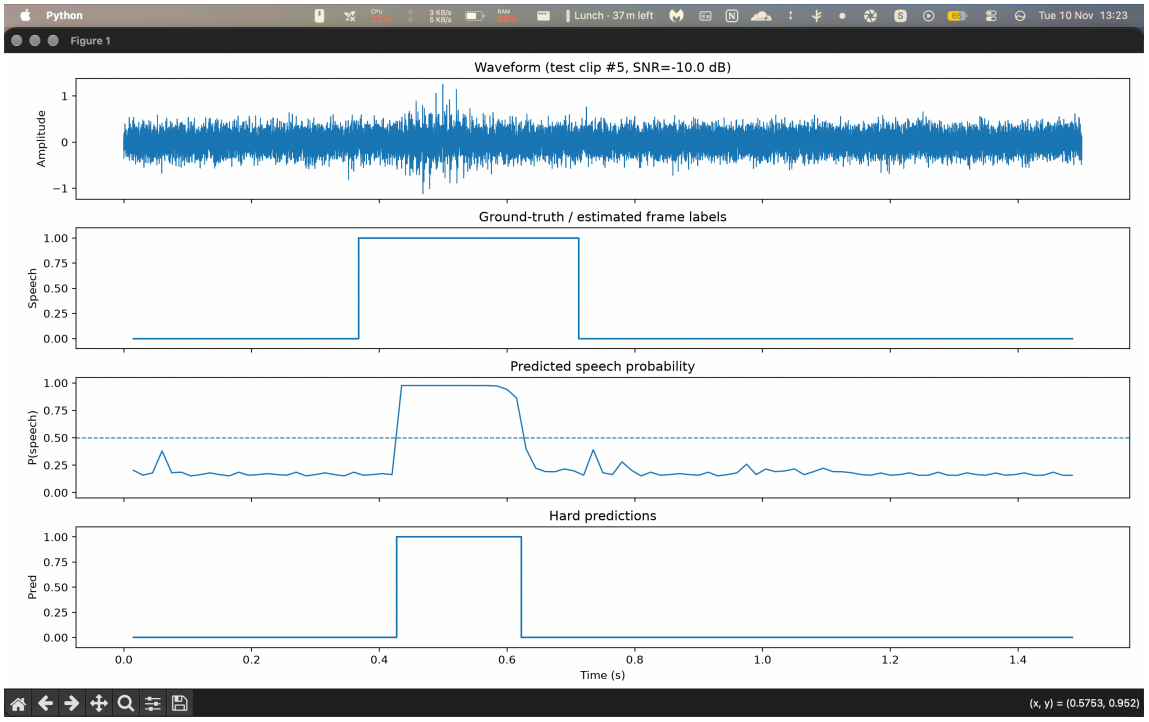
▼ Stage 3 Findings

▼ Results

Prediction for test clip 5, Noise = 0dB



Prediction for test clip 5, Noise = -10dB



Observation	Interpretation	Results
Predictions closely follow labels	Successful detection	Eh
Prediction begins late	Possible onset delay	Yes
Prediction ends late	Possible offset delay	Dependent on noise level
Speech is missed entirely	False negatives	No
Noise is classified as speech	False positives	No
Prediction flickers rapidly	Temporal instability	No

Stage: 3 — Real-speech pilot, noise-robustness evaluation, and qualitative visualization

Status: Completed as a pilot; original-paper reproduction remains open

Reference: Yang et al., *sVAD: A Robust, Low-Power, and Light-Weight Voice Activity Detection with Spiking Neural Networks*, ICASSP 2024

Next stage: Stage 4 — Architectural comparison, beginning with diagonal versus dense recurrence

1. Objective

Move from Stage 2's synthetic tone-and-silence debugging task to real spoken-word recordings, measure frame-level speech/non-speech detection under multiple signal-to-noise ratios (SNRs), and inspect the model's temporal errors visually. Preserve the Stage 2 architecture for a controlled baseline.

2. Dataset and evaluation scope

- **Dataset:** A pilot subset of Google Speech Commands v0.02, prepared with **180 training, 50 validation, and 50 test recordings** (as configured in the preparation command).
- **Audio:** 16 kHz; 30 ms analysis frames with 15 ms hop (50% overlap); model outputs a binary speech decision per frame.
- **Annotations:** Frame-level speech intervals were estimated from signal energy during preparation. They are **pseudo-labels**, not manually annotated VAD ground truth. HTER therefore measures agreement with these provisional labels.
- **Noise:** Controlled additive noise, described in the pilot as Gaussian; not the original paper's recorded QUT-NOISE-TIMIT environments.
- **Evaluation:** HTER = (MR + FAR)/2, where MR = FN/(TP+FN) and FAR = FP/(FP+TN). Lower is better. Six test conditions: −10, −5, 0, +5, +10, +15 dB SNR.
- **Test volume:** Across each SNR condition, evaluation reported **1,654 labelled speech frames and 3,296 labelled non-speech frames** (4,950 total).

Methodological limitation: The experiments are a real-speech *pilot*, not a faithful reproduction of the original paper's dataset, acoustic conditions, or annotation protocol. The independent test set has been consulted for exploratory comparisons; future hyperparameter and checkpoint selection should be done on validation data only.

3. Baseline model and training settings

We carried forward the working **4,313-parameter** sVAD reconstruction:

Component	Trainable parameters
SincNet + spiking 1D convolution encoder	1,260
Spiking attention block	2,283
Diagonal-recurrent spiking classifier + readout	770
Total	**4,313**

Classifier: 20-dimensional encoded features → 32 recurrent hidden neurons → 2 readout logits. LIF settings: decay $\alpha = 0.5$, firing threshold $\theta = 0.3$; surrogate parameter $a = 4$. Architecture-specific assumptions, including

101-sample SincNet filter, convolution kernel size 3, attention widths 31/31, and diagonal recurrence, are **not confirmed as the authors' exact design**.

The pilot used class-weighted cross-entropy and **λ = 0** (attention MSE disabled in the loss). The reported initial training counts were 11,195 non-speech and 6,625 speech frames, giving class weights approximately **[0.7959, 1.3449]** for [non-speech, speech]. Training ran on the Mac's **MPS** backend. Checkpointing selected the best validation HTER.

4. Experiment A — Single-SNR training, 20 epochs

Training with noise at **0 dB SNR** gave a best validation HTER of **13.4%** at epoch 20 (MR = 20.7%, FAR = 6.2%). The training loss decreased from approximately 0.699 to 0.343. The checkpoint was saved as `runs/stage3_real_best.pt`.

Held-out test SNR sweep

SNR (dB)	MR (%)	FAR (%)	HTER (%)
−10	35.31	7.01	21.16
−5	29.69	3.43	16.56
0	22.73	4.70	**13.72**
+5	16.87	14.75	15.81
+10	11.61	30.19	20.90
+15	7.74	50.39	29.07

Observation: The model's false-alarm rate rose steeply at high SNR, while miss rate fell. This suggests sensitivity to the training noise distribution; it is not proof of a specific internal mechanism.

5. Experiment B — Multi-SNR training, 20 epochs

The model was newly trained with sampled SNRs **{−10, −5, 0, +5, +10, +15} dB**, retaining architecture and λ = 0.

SNR (dB)	MR (%)	FAR (%)	HTER (%)
−10	39.36	3.94	21.65
−5	31.08	4.28	17.68
0	26.12	3.73	14.93
+5	21.58	6.07	13.83
+10	18.50	9.50	14.00
+15	16.75	13.80	15.28

Unweighted mean across SNR conditions: approximately **16.23% HTER**, versus **19.54%** for single-SNR training. High-SNR performance became much more consistent, though results were slightly worse at the three lowest SNRs. These are exploratory comparisons, not statistical significance tests.

6. Experiment C — Multi-SNR training, 60 epochs

A **fresh 60-epoch training run** was performed with the same six-SNR training range. The best validation HTER reported at completion was **0.135299… (13.53%)**, stored in `runs/stage3_multi_snr_60_best.pt`. The logged training loss at epoch 60 was approximately **0.3326** (versus approximately 0.701 at the beginning of the first real-speech experiment). The saved-best epoch was not independently established in the conversation; the run logged improvements through epochs 55–56.

Corrected 60-epoch checkpoint evaluation

SNR (dB)	MR (%)	FAR (%)	HTER (%)
−10	33.37	7.37	20.37
−5	29.75	4.00	16.88

SNR (dB)	MR (%)	FAR (%)	HTER (%)
0	25.33	3.46	14.40
+5	22.31	2.97	12.64
+10	19.29	3.64	**11.46**
+15	17.29	5.76	11.53

The six-SNR mean HTER decreased from **16.23% (20-epoch multi-SNR)** to **14.55% (60-epoch multi-SNR)**—an improvement of approximately **1.68 percentage points**. The 60-epoch model improved HTER at **all six** tested SNRs, with a relatively low false-alarm rate across the sweep. Its worst condition remained −10 dB, HTER **20.37%**.

Checkpoint-evaluation troubleshooting

Initial evaluation after the 60-epoch run appeared to reproduce the 20-epoch results exactly. Inspection of the command showed that `--checkpoint runs/stage3_multi_snr_best.pt` (the **old** checkpoint) had been passed instead of the new `_60_best.pt` checkpoint. The model checkpoints had distinct state dictionaries (summed absolute weight difference ≈ **189.2934**). Re-evaluating the correct checkpoint produced the different 60-epoch results above. This was a command/checkpoint-selection mistake, not an established model-loading defect.

7. Experiment D — Frame-level visualization

A visualization script was added to display, for a chosen test clip and SNR:

1. Noisy waveform;
2. Energy-derived speech/non-speech reference labels;
3. Predicted speech probability with 0.5 decision threshold;
4. Binary predicted speech regions.

Several integration issues were resolved while adapting it to the actual Stage 3 manifest and data-loader interfaces:

- Use `clean_path` rather than the assumed `audio_path`;
- Read `speech_intervals` rather than `speech_regions`;
- Obtain sample rate from the model config rather than a nonexistent `sample_rate` manifest key;
- Resolve manifest-relative audio paths against the manifest's parent directory;
- Call `frame_labels(intervals, n_samples, sample_rate, frame_samples, hop_samples, threshold=0.5)` using **sample counts**, not the mistaken keyword `total_samples`.

Visual findings: On a representative clip at 0 dB, speech probability rose to approximately 0.95–0.98 during the estimated speech segment and stayed around 0.15–0.20 outside it. Hard predictions broadly matched the annotated speech region. Across additional manual visual checks, the user reported a **consistent apparent onset delay**, with **offset behavior varying by noise condition**.

These are **qualitative observations**, not measured onset/offset latency statistics. The delay may reflect SNN temporal integration, framing/timestamp conventions, or errors in automatically inferred speech boundaries. No causal attribution has been confirmed.

8. Consolidated comparison

Model/training	Mean test HTER over six SNRs	Worst test HTER	Best observed test SNR
Single-SNR, 20 epochs	19.54%	29.07%	0 dB (13.72%)
Multi-SNR, 20 epochs	16.23%	21.65%	+5 dB (13.83%)
Multi-SNR, 60 epochs	**14.55%**	**20.37%**	+10 dB (11.46%)

Primary result: Expanding training noise diversity mitigated the severe high-SNR false-alarm issue, and extending training from 20 to 60 epochs further improved test HTER across every evaluated noise condition.

9. Confirmed findings versus open questions

Supported by completed experiments

- The 4,313-parameter reconstruction can train on real spoken-word recordings and produce meaningful frame-level speech probabilities.
- The best 20-epoch single-SNR model reached 13.4% validation HTER at 0 dB in the pilot setup.
- Multi-SNR training improved average test robustness relative to single-SNR training.
- A 60-epoch multi-SNR run reached 14.55% mean test HTER across six conditions, with 20.37% at −10 dB.
- The visualization tool operates with the pilot dataset after adapting paths and annotations; example clips show plausible detections.

Still unresolved

- Real speech boundary accuracy: pseudo-labels have not been validated against hand-labelled onset/offset times.
- Generalizability beyond a small spoken-word pilot: no QUT-NOISE-TIMIT or manually annotated VAD benchmark yet.
- Whether offset variation and onset delays stem primarily from architecture, frame conventions, noise, or pseudo-labels.
- Original-paper architecture: exact SincNet/sConv/attention parameters and recurrent connectivity remain unknown.
- Whether the noise-generation/evaluation protocol is completely comparable between runs; future comparisons should fix identical noisy test inputs and random seeds explicitly.
- The attention-reconstruction objective was inactive ($\lambda = 0$) in the successful pilot; the full paper's attention training objective has not yet been reproduced.

10. Stage 4 plan — Controlled recurrence comparison

Research question: Under otherwise matched training and test conditions, is **diagonal recurrence** or **dense recurrence** more effective for the sVAD classifier, and how do they differ in compute/parameter cost and onset/offset behavior?

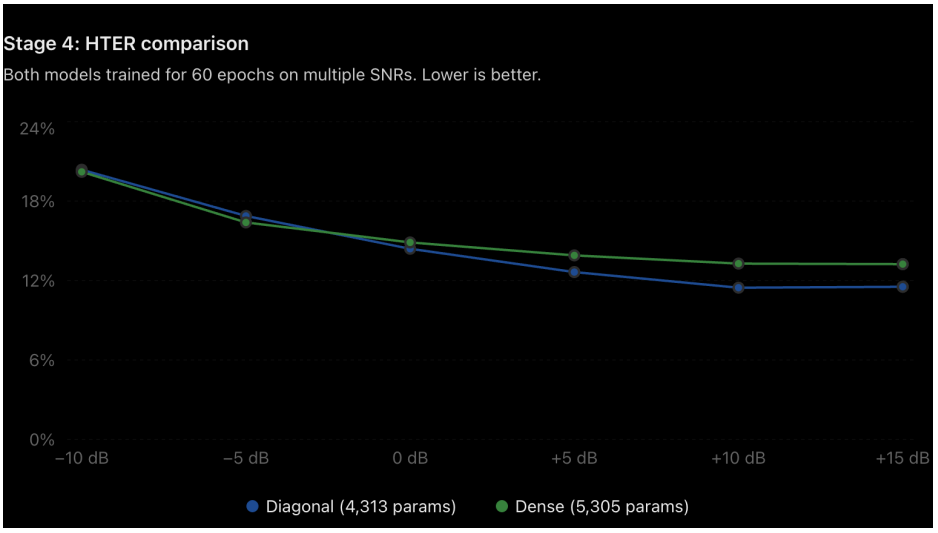
1. Preserve the current diagonal-recurrence checkpoint as the baseline.
2. Train a **dense-recurrence** candidate with the same encoder, attention configuration, dataset split, noise conditions, class-weighting procedure, optimizer, seed(s), epoch budget, and checkpoint-selection rule. In this direct comparison, dense recurrence increases classifier count from **770 to 1,762**, making total model size **5,305** parameters.
3. Evaluate on validation data across the same six SNRs, then report fixed held-out test results. Keep test noise samples identical across candidates.
4. Compare MR, FAR, HTER per SNR, number of parameters, and average spike activity; examine onset/offset differences on the same clips.
5. If a **parameter-matched** comparison is desired, perform a second experiment changing attention or convolution widths to bring dense recurrence back near 4.3K, while acknowledging that changing other layers introduces an additional variable.
6. Once the recurrence question is resolved provisionally, revisit $\lambda > 0$ and more realistic noise/ground-truth VAD annotations.

Stage gate: Stage 3 is closed as a successful real-speech *pilot*, not as final reproduction of the original paper. Stage 4 begins with the diagonal-versus-dense experiment, followed by quantitative timing analysis.

▼ Stage 4 Findings

Over a small synthetic database,

SNR	Diagonal HTER	Dense HTER	Difference (Dense – Diagonal)
−10 dB	20.37%	20.21%	−0.16 pp
−5 dB	16.88%	16.39%	−0.49 pp
0 dB	14.40%	14.88%	+0.48 pp
+5 dB	12.64%	13.90%	+1.26 pp
+10 dB	11.46%	13.28%	+1.82 pp
+15 dB	11.53%	13.24%	+1.71 pp
Mean	14.55%	15.32%	+0.77 pp



The three setups to compare:

- Diagonal, 4,313 parameters: Our established baseline.
- Dense, 5,305 parameters: Tests the effect of adding recurrent connectivity without reducing attention capacity.
- Matched Dense, 4,311 parameters: Tests whether allocating more of the same parameter budget to recurrence improves performance.

Using random seeds to check the difference between a matched dense vs diagonal setup.

Stage: 4 — Recurrent Architecture Investigation

Status: Completed (exploratory comparison)

Date: 10 November 2026

Source: Experiments and terminal outputs supplied during this project; results are measured against the Stage 3 real-speech pilot's automatically estimated frame labels, *not* the paper's QUT-NOISE-TIMIT annotations.

1. Objective

Investigate whether the reconstructed sVAD classifier should use diagonal/self-recurrence or dense/full recurrence, while assessing model-size trade-offs and reproducibility. This stage does **not** establish the authors' original recurrence implementation.

2. Experimental setup

- Dataset: Stage 3 Google Speech Commands v0.02 pilot, with energy-derived speech intervals; held-out test split contains 1,654 speech and 3,296 non-speech evaluated frames.
- Input: 16 kHz, 30 ms frames and 15 ms hops; 20 encoded features.
- Classifier: 32 spiking recurrent hidden neurons and 2 output units.

- Training: 60 epochs; batch size 8; 1.5 s segments; training SNRs −10, −5, 0, +5, +10, +15 dB; class-weighted cross-entropy; attention-reconstruction loss weight $\lambda = 0$.
- Test metric: HTER = (miss rate + false alarm rate)/2; reported values are percentages.
- Noise: pilot's generated Gaussian noise, not the original paper's recorded environmental noises.
- Caveat: reported checkpoints were selected with the existing validation procedure rather than a fully multi-SNR validation selection criterion. Some architecture decisions were informed by earlier test inspections, so the test split is no longer an untouched final benchmark.

3. Architectures

Component	Diagonal baseline	Full-size dense	Parameter-matched dense
Encoder parameters	1,260	1,260	1,260
Attention hidden widths	31, 31	31, 31	17, 24
Attention parameters	2,283	2,283	1,289
Classifier parameters	770	1,762	1,762
Total parameters	**4,313**	**5,305**	**4,311**

The dense recurrent matrix uses 32 × 32 = 1,024 weights, versus 32 self-recurrent weights for diagonal recurrence, accounting for 992 extra recurrent parameters. The matched dense architecture reduces attention width to offset those extra recurrent weights. Therefore the matched comparison changes *both* recurrence structure and attention capacity.

4. Initial seed-42 architecture comparison

SNR	Diagonal 4,313	Full-size dense 5,305	Matched dense 4,311
−10 dB	20.37%	20.21%	**19.14%**
−5 dB	16.88%	16.39%	**16.20%**
0 dB	14.40%	14.88%	**13.73%**
+5 dB	12.64%	13.90%	**11.78%**
+10 dB	11.46%	13.28%	**11.09%**
+15 dB	**11.53%**	13.24%	11.56%
Six-SNR mean	**14.55%**	**15.32%**	**13.92%**

The larger dense model did not outperform the diagonal model overall in the seed-42 experiment. The parameter-matched dense configuration did outperform it overall, motivating replication.

5. Three-seed repeatability experiment

The diagonal and matched-dense architectures were independently trained at seeds 42, 123, and 2026. The seed does not transfer trained weights between architectures.

5.1 Mean HTER per run

Seed	Diagonal (4,313)	Matched dense (4,311)	Lower HTER
42	14.55%	**13.92%**	Matched dense
123	**14.33%**	14.67%	Diagonal
2026	**13.88%**	14.11%	Diagonal
Mean across three seeds	14.25%	**14.23%**	Effectively tied
Sample standard deviation across seed means	0.34 percentage points	0.39 percentage points	—

The paired dense-minus-diagonal six-SNR mean differences were −0.63, +0.34 and +0.24 percentage points for seeds 42, 123 and 2026. Their average is **−0.018 percentage points**, too small relative to run variability to claim an overall winner.

5.2 HTER by SNR, averaged across three seeds

SNR	Diagonal	Matched dense	Difference: dense − diagonal
−10 dB	20.64%	**18.98%**	−1.66 pp
−5 dB	16.50%	**16.36%**	−0.14 pp
0 dB	14.03%	**13.97%**	−0.06 pp
+5 dB	**12.05%**	12.31%	+0.26 pp
+10 dB	**11.06%**	11.69%	+0.62 pp
+15 dB	**11.21%**	12.08%	+0.87 pp

At −10 dB, matched dense beat diagonal in all three seeds, suggesting a possible advantage in difficult noise. At +10 and +15 dB, diagonal beat matched dense in all three seeds. These are **pilot observations**, not evidence of statistical significance or proof that recurrence alone caused the differences.

6. Qualitative waveform review

Visual inspection of selected recordings showed a recurrent positive onset delay relative to energy-derived reference labels. At −10 dB, predictions sometimes fragmented or terminated early; at 0 dB, one selected clip showed a continuous, well-aligned speech prediction. Speech offset behavior appeared noise-dependent. These observations are anecdotal and need systematic timing measurements with carefully aligned frame timestamps and trustworthy annotation boundaries.

7. Interpretation and caveats

Supported by experiments: Both recurrence choices train and achieve similar overall mean HTER within a near-identical parameter budget; changing parameter allocation can affect performance; the matched dense candidate was better at −10 dB across these three seed runs.

Not established: Which recurrence architecture the paper's authors used; an intrinsic advantage from dense recurrence (attention was also changed); accuracy against manually annotated speech boundaries; reproducibility on QUT-NOISE-TIMIT; hardware power consumption or latency; statistical significance from three seeds.

Additional caution: fixed evaluation noise realizations and a larger, fully separate evaluation set would strengthen inference. The test split has been repeatedly inspected throughout tuning, so final claims should be assessed on a new untouched set.

8. Stage 4 conclusion

The parameter-matched dense and diagonal designs are **essentially tied in overall accuracy** in this pilot, at **14.23% vs 14.25%** three-seed mean HTER. Diagonal recurrence remains a sensible lightweight reference, because its classifier needs 992 fewer recurrent weights. Retain matched dense as a low-SNR alternative and avoid claiming the original architecture has been identified.

9. Proposed Stage 5

1. Prioritize remaining unspecified architecture choices: internal SNN timesteps, SincNet filter bank and kernel settings, sConv1D dimensions, attention mask semantics, and the paper's λ.
2. Implement proper multi-SNR validation-based model selection and a fresh held-out evaluation protocol. Use matched noise realizations for comparisons.
3. Restore and sweep nonzero λ with clean/noisy paired audio, since Stage 3–4 experiments intentionally used λ = 0 and therefore did not test the central attention-reconstruction training objective.
4. Add quantitatively measured speech onset/offset errors and inspect failure types.
5. Transition toward manually validated VAD boundaries and, where access permits, the QUT-NOISE-TIMIT noise environments and train/test protocol.

6. Only after accuracy is reliable, assess model footprint, spike rates and energy *estimates*, explicitly distinguishing simulated estimates from measured hardware power.

10. Overall project status

Stage	Goal	Status
1	Initial paper-based architecture and parameter-count reconstruction	Complete as initial implementation; architecture still uncertain
2	Synthetic-audio pipeline, learning/debug checks, report	Complete
3	Real-speech pilot, multi-SNR training, qualitative visualization, report	Complete as pilot
4	Recurrent architecture comparisons and three-seed replication	Complete as exploratory study
5	Remaining architecture and training assumptions, faithful benchmark, robustness and efficiency validation	Not started

Overall assessment: Four project stages are complete at the exploratory/prototyping level. This is a working, tested reconstruction, **not yet a faithful reproduction of the ICASSP paper**. The largest outstanding scientific gaps are verified architecture specifics, the original joint attention loss (λ > 0), stronger reference labels, original dataset protocol, and rigorous power/latency evaluation.

> 🔒 **Important Note:**
> When training a diagonal model then training a dense model, we actually use a fresh model for both runs, so it's not that the same model has been trained twice, favouring the training that was done later.

▼ Stage 5 Findings

Note that

$$\mathrm{SNR}_{dB} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right)$$

Stage: 5 — Clean-reference attention reconstruction loss and gradient treatment

Status: Completed as an exploratory pilot; no reproducible accuracy benefit established

Report date: 12 November 2026

Evidence: Model outputs, validation metrics, diagnostic readings and three-seed reproducibility summaries supplied during the project. All values are from the user's local runs, not independently rerun here.

1. Research question

Stages 3–4 trained sVAD using weighted cross-entropy (CE) with attention reconstruction weight λ = 0. Stage 5 asked whether including a clean-reference feature reconstruction objective improves noisy speech detection, and whether stopping reconstruction-loss gradients through the clean-reference branch changes performance or training dynamics.

Loss: **L = L_CE + λ L_MSE**, where **L_MSE = mean[(attended noisy features − clean reference features)²]**. Clean reference inputs are the original dataset recordings, not guaranteed noise-free. No extra Gaussian noise is injected into this reference branch. The same trainable encoder processes clean and noisy signals.

Two gradient approaches:

- **Joint:** reconstruction-loss gradients affect the clean and noisy encoder paths.
- **Detached:** clean features are detached for reconstruction loss, so its gradients do not flow through the clean branch. This does not globally freeze encoder parameters.

2. Setup and constraints

- Dataset: Google Speech Commands v0.02 pilot, with energy-estimated speech intervals, not manually verified VAD annotation.
- 16 kHz; 30-ms frames with 15-ms hop; 20 features; diagonal spiking recurrent classifier with 32 hidden neurons and 2 output neurons.
- Reconstructed 4,313-parameter encoder + attention + diagonal-recurrent architecture; original authors' precise parameterization remains unverified.
- Noise: added Gaussian noise at −10, −5, 0, +5, +10, +15 dB; not the original QUT-NOISE-TIMIT acoustic noise distribution.
- Training: 60 epochs maximum; batch size 8; 1.5-second segments; weighted CE; Adam initial learning rate 0.001; step schedule every 40 epochs, ×0.1.
- Checkpoints selected by **mean validation HTER across the six SNRs**; HTER = (miss rate + false alarm rate) / 2. Individual saved best epochs may be lower than 60.
- Stage 5 validation evaluated 1,801 speech frames and 3,149 nonspeech frames per SNR for the seed-42 diagnostic data.
- A few prior test-set comparisons informed the project, meaning the historical test split is **not entirely pristine** for final claims.

3. Single-seed coefficient and gradient sweep (seed 42)

ID	λ	Clean-target gradient	Saved epoch	Mean validation HTER
A	0	Joint (irrelevant at λ=0)	60	15.216%
D	0.01	Detached	51	14.569%
B	0.1	Detached	57	14.741%
E	0.5	Detached	49	14.680%
F	0.01	Joint	60	14.745%
C	0.1	Joint	59	**14.502%**
G	0.5	Joint	56	15.304%

Interpretation: the best seed-42 validation score was joint λ=0.1; the detached configuration showed less sensitivity to the tested λ values. These were exploratory configuration choices, not independent confirmation of a general advantage.

3.1 Validation HTER by SNR, seed 42 (percent)

Configuration	−10	−5	0	+5	+10	+15
A: baseline λ=0	**21.75**	17.37	13.81	12.81	12.33	13.23
D: detached λ=0.01	22.63	17.65	13.69	11.46	10.71	11.27
B: detached λ=0.1	22.22	18.05	13.66	11.59	11.38	11.54
E: detached λ=0.5	23.14	17.72	**13.07**	**11.13**	11.10	11.91
F: joint λ=0.01	22.71	17.73	13.30	11.52	11.18	12.03
C: joint λ=0.1	21.97	**17.10**	13.75	11.31	**10.93**	11.96
G: joint λ=0.5	22.63	18.38	14.88	12.92	11.82	**11.20**

All nonzero-λ candidates had **higher HTER at −10 dB** than the λ=0 baseline in this seed-42 validation sweep. Gains in six-SNR means therefore largely reflect other SNR ranges.

4. Representation and attention diagnostics

The diagnostic measured CE, unweighted feature MSE, encoder spike rate, attention-mask active fraction, post-attention active fraction, and frames whose attention mask was entirely zero. These quantities are descriptive, not independent measures of feature usefulness.

4.1 Representative diagnostic at 0 dB

Model	CE	MSE	Encoder rate	Mask rate	Attended rate	All-zero mask frames
Baseline λ=0	0.3231	0.1586	0.351	0.439	0.154	0.000
Detached λ=0.01	0.3242	0.1216	0.321	0.536	0.230	0.000
Detached λ=0.1	0.3238	0.1121	0.319	0.594	0.275	0.000
Detached λ=0.5	0.3190	0.0738	0.253	0.543	0.204	0.000
Joint λ=0.01	0.3117	0.1146	0.265	0.531	0.198	0.000
Joint λ=0.1	0.3251	0.0623	0.212	0.498	0.193	0.001
Joint λ=0.5	0.3417	**0.0136**	**0.144**	**0.398**	**0.131**	0.000

The joint λ=0.5 model lowers the unweighted MSE from 0.0623 (joint λ=0.1) to 0.0136 at 0 dB but worsens CE and HTER, with encoder activity dropping from 21.2% to 14.4%. This is consistent with, but **does not prove**, a representation-suppression or reduced-activity shortcut. Spiking sparsity can also arise from useful feature selection. Near-zero all-zero-mask frequency rules out frequent *complete mask shutdown*, not partial information loss.

4.2 Selected high- and low-SNR observations

- At −10 dB, attended-feature spike rates: baseline 0.126, detached λ=0.1 0.216, joint λ=0.1 0.157; despite the activity differences the baseline had the best HTER among those three.
- At +15 dB, corresponding rates: 0.194, 0.308, 0.214; detached λ=0.1 achieved HTER 11.54% versus baseline 13.23% on validation.
- At −10 dB, detached λ=0.5 had all-zero mask frames 0.006 (0.6%) and HTER 23.14%; this is an observation only, not evidence of causality.
- Higher attended activation is not intrinsically better: masks may pass noise as well as speech cues.

5. Cross-seed reproducibility experiment

After the exploratory sweep, three configurations were independently trained from scratch at seeds 42, 123 and 2026:

1. λ=0, baseline;
2. detached λ=0.01;
3. joint λ=0.1.

Configuration	Seed 42 HTER	Seed 123 HTER	Seed 2026 HTER	Three-seed mean	Sample SD
Baseline λ=0	15.22%	14.33%	14.62%	14.72%	0.45 pp
Detached λ=0.01	14.57%	14.78%	14.68%	**14.67%**	**0.10 pp**
Joint λ=0.1	14.50%	14.51%	15.16%	14.72%	0.38 pp

These are rounded summaries shared by the user. In comparisons or code, use full-precision checkpoint values when available.

5.1 Paired improvements over baseline

Baseline HTER minus experimental HTER (positive means improvement).

Seed	Detached λ=0.01	Joint λ=0.1
42	+0.65 pp	+0.71 pp
123	−0.45 pp	−0.18 pp
2026	−0.06 pp	−0.54 pp
Mean paired difference	**+0.05 pp**	**~0.00 pp**

Both alternatives improved on seed 42 but worsened on seeds 123 and 2026. Therefore the initial seed-42 accuracy advantage **did not replicate consistently**. Detached λ=0.01 exhibited the lowest *observed* between-seed SD; only three seeds were tested, so improved training stability remains a hypothesis.

6. Research conclusions

1. **No reproducible classification accuracy advantage was established** for the tested clean-reference MSE loss. The detached approach's ~0.05 pp mean advantage is small compared with between-seed variability.
2. Joint and detached MSE gradients measurably affect encoder and attention activity.
3. Larger joint λ (0.5) reduced unweighted feature MSE substantially and lowered encoder spike activity, but did not produce better mean validation HTER.
4. Across all seven seed-42 configurations, none improved validation HTER at −10 dB over the cross-entropy-only baseline.
5. These findings **do not disprove** the original paper's architecture or attention claims: the reconstructed details and evaluation protocol differ from the paper's original benchmark.

7. Limitations and experimental integrity

- Energy-estimated speech segments are proxy VAD labels; clean recordings can already contain ambient noise.
- Gaussian-noise corruption does not emulate all real noise environments.
- Only three seeds for final comparison; no high-confidence statistical significance claim.
- The parameter count alone does not establish a faithful reconstruction of the original SNN architecture.
- Seed 42 served for exploratory hyperparameter selection, so the multi-seed analysis is replication/robustness evidence rather than a preregistered test.
- The prior test split was inspected during earlier stages; use a newly held-out dataset for an eventual final generalization claim.
- MSE diagnostic values can decline partly because the target features themselves change with joint gradients, and should not be treated as direct measures of speech information recovery.
- Checkpoint key names may differ (`val_metrics`, etc.); presence/absence of a particular metadata key does not alone show that a checkpoint is unusable.

8. Stage 6 handoff: attention ablation

Research question: Does the attention block itself contribute to noisy speech detection when λ=0, independent of the reconstruction-loss objective?

Design:

- Compare (A) full encoder → learned attention mask → sRNN versus (B) same encoder → all-ones pass-through mask → sRNN.
- Fix λ=0 in both, with the same frame settings, dataset manifest, training recipe, six SNRs, model-selection criterion and matched seeds (42, 123, 2026).
- Train all ablated models **from scratch**. Merely bypassing attention in a pretrained model is a separate inference-only test with a distribution-shift confound.

- Count trainable parameters: full model = 4,313; bypass model = 2,030 (attention's 2,283 weights absent). Thus this ablation estimates the benefit of the complete attention block, **not** a parameter-matched architectural effect.
- Report mean validation HTER, paired differences by seed, and per-SNR MR/FAR/HTER; consider extra comparisons for parameter control only afterward.
- Keep all training and model-selection decisions on validation data; reserve any fresh held-out evaluation for a later confirmatory stage.

9. Status

Stage 5 research question answered to the extent supported by pilot data; Stage 6 attention-block ablation is the next experiment. The overall project remains an experimental reconstruction, not a verified replication of the ICASSP 2024 sVAD implementation or hardware power evaluation.

▼ Stage 6 Findings

Study: Ablation of the learned spiking attention module

Status: Completed exploratory, three-seed functional ablation

Evaluation: Validation split, Speech Commands pilot, artificial Gaussian noise at −10, −5, 0, +5, +10 and +15 dB SNR

Date documented: 12 November 2026

1. Research question

Does the spiking attention module improve frame-level voice activity detection in our reconstructed sVAD network relative to directly passing the encoder features to the recurrent classifier?

Stage 5 established that auxiliary clean-feature reconstruction loss did **not** show a dependable accuracy benefit across three seeds. We therefore compared architectures trained with classification loss alone ($\lambda = 0$) to isolate the *presence* of learned attention rather than the effect of the reconstruction objective.

2. Experimental design

We compared two models:

Component	Full attention	Attention bypass
SincNet + spiking-convolution encoder	Present	Present
Learned spiking attention mask	Present	Removed/bypassed
Feature passed to recurrent classifier	Attended noisy features $\hat{S} = Y \odot M$	Encoder features $\hat{S} = Y$
Diagonal spiking recurrent classifier	Present	Present
Encoder parameters	1,260	1,260
Attention parameters	2,283	0
Classifier parameters	770	770
Total parameters	**4,313**	**2,030**

Training: 60 epochs per experiment; independent fresh initialization per run; seeds **42, 123, 2026**; batch size **8**; audio segment length **1.5 s**; training and checkpoint-selection validation SNRs **−10, −5, 0, +5, +10, +15 dB**; checkpoints selected by the lowest mean validation half-total error rate (HTER) across six SNRs. Both models used $\lambda = 0$. Dataset: Speech Commands pilot with energy-derived pseudo speech intervals, so labels are not manually adjudicated speech activity annotations. Noise was synthetic Gaussian noise, not the recorded environmental noise of the original paper.

Metrics: miss rate MR = FN/(TP+FN), false alarm rate FAR = FP/(TN+FP), and HTER = (MR + FAR)/2. Lower HTER is better.

Causal interpretation caveat: This is a functional attention ablation, but not a capacity-matched ablation. Removing attention reduces the trainable parameter count by 2,283 (52.9% of the full model). Therefore performance differences cannot be attributed solely to the attention computation independent of extra model capacity.

3. Per-seed validation results

Seed	Full checkpoint best epoch	Full HTER	Bypass checkpoint best epoch	Bypass HTER	Bypass − full (pp)
42	60	14.731%	51	14.661%	−0.069
123	55	14.393%	56	14.764%	+0.372
2026	60	14.864%	50	15.452%	+0.588
Mean	—	**14.662%**	—	**14.959%**	**+0.297**
Sample SD	—	**0.243 pp**	—	**0.430 pp**	—

Positive values in the final column favor learned attention. The best checkpoint's epoch is not necessarily the final training epoch, since checkpoint selection uses validation performance.

Result: The full model outperformed bypass in **2 of 3 seeds**, with a **0.297-percentage-point mean improvement** (about **2.0% relative** to the bypass mean HTER). Seed 42 slightly favored bypass. The full model showed lower observed between-seed variation, though three seeds are insufficient to confirm a general stability advantage.

4. Per-SNR validation results, averaged over matched seeds

SNR	Full HTER	Bypass HTER	Bypass − full (pp)
−10 dB	22.54%	22.95%	+0.41
−5 dB	17.33%	17.79%	+0.45
0 dB	13.31%	13.56%	+0.25
+5 dB	11.28%	11.80%	+0.52
+10 dB	11.17%	11.42%	+0.25
+15 dB	12.35%	12.25%	−0.10

Full attention performed better at **5 of 6** SNRs. It improved the noisiest cases by 0.41 pp (−10 dB) and 0.45 pp (−5 dB), but the largest observed improvement occurred at +5 dB (0.52 pp). Thus the pattern is compatible with an attention benefit under noise, but does **not** demonstrate that attention specifically helps low-SNR cases more than other cases.

The per-SNR figures above were reported rounded to two decimal places, and may not aggregate to exactly the displayed across-SNR mean because of rounding.

5. Parameter-efficiency interpretation

Quantity	Result
Full model parameter count	4,313
Bypass parameter count	2,030
Removed parameters	2,283
Fraction of full parameters removed	52.9%
Full model parameter increase relative to bypass	112.5%
Mean HTER gain with full attention	0.297 percentage points

The bypass network sacrifices a relatively small amount of mean validation performance while using fewer than half as many parameters. This is relevant to the **lightweight** goal, but parameter counts alone cannot establish energy cost, computational latency, power consumption, or hardware efficiency. Those require appropriate instrumentation, spike/operation counts, or a hardware implementation.

6. Conclusions and limitations

1. Within this reconstructed architecture and pilot dataset, learned spiking attention yielded a **modest average validation HTER improvement** over simply bypassing attention.
2. The result is **not uniform across seeds**: attention won in two runs and lost slightly in one.
3. A separate Stage 5 result shows that **attention can help in the absence of an additional MSE objective**, while the added reconstruction loss did not show a reliable three-seed classification gain.
4. The ablation **does not isolate the inductive bias of attention from parameter budget**, because the networks have 4,313 versus 2,030 trainable parameters.
5. Training/validation noise was synthetic Gaussian; the dataset contains energy-derived pseudo-boundaries, not manually verified VAD labels; and only three random seeds were tested. The conclusions do not constitute a replication of original sVAD paper results on QUT-NOISE-TIMIT.
6. These are validation data used during model selection, not an independent final evaluation. A held-out assessment and careful prevention of test-driven hyperparameter choices remain necessary.

Stage 6 conclusion: The full attention model performed modestly better on average (14.662% vs 14.959% validation HTER) but at substantially higher model capacity. A parameter-matched no-attention control is the logical next ablation.

7. Handoff to Stage 7

Question: When model capacity is approximately fixed at 4.3K parameters, does learned attention improve VAD performance compared with spending the parameter budget elsewhere?

Proposed controls: Reuse the Stage 6 full-attention model; add a no-attention model with a larger recurrent classifier to achieve approximately the same parameter count; repeat matched-seed training and six-SNR validation. Varying the recurrent hidden width and recurrent connectivity changes more than parameter budget, so interpret a result as an architecture-budget comparison, not a perfectly isolated causal test of attention.

Reproduction requirements: Same data, splits, noise draws or evaluation noise seeds, preprocessing, frame labels, optimizer/schedule, epoch budget, checkpoint selection rule, and seeds across configurations. Train new capacity-matched models from scratch; do not resume Stage 6 models. Preserve all prior checkpoints and logs.

▼ Stage 7 Findings

Our dense recurrent classifier has the parameter formula:

$$N_{\text{classifier}} = H^2 + 23H + 2$$

where H is the number of hidden neurons.

For \(H=45\):

$$N_{\text{classifier}} = 45^2 + 23(45) + 2 = 3062$$

Adding our 1,260-parameter encoder:

$$N_{\text{total}} = 1260 + 3062 = \boxed{4322}$$

Research question

Does the learned spiking attention architecture retain an advantage when compared against a no-attention model with approximately the same parameter budget?

Configurations

Arm	Attention mode	Recurrent mode	Hidden neurons	Total parameters
A (full)	full	diagonal	32	4,313
B (lightweight reference)	bypass	diagonal	32	2,030
C (capacity-matched control)	bypass	dense	45	4,322

The parameter counts use the established implementation: encoder = 1,260; full attention = 2,283; diagonal hidden-32 recurrent classifier = 770. For a dense recurrent classifier with input size 20, hidden width H, output size 2, and both biases enabled:

`classifier_parameters = (20*H + H) + (H*H) + (2*H + 2) = H² + 23H + 2`.

At `H=45`, classifier = 3,062, so bypass total = `1,260 + 3,062 = 4,322` (only nine more than the 4,313-parameter attention model). **Verify by counting the instantiated model's trainable parameters**; the formula assumes the same bias and recurrence definitions used in earlier stages.

Scientific limitation

This matches *total parameter count*, not architecture. Arm C differs in recurrent connectivity (dense vs diagonal) and width (45 vs 32). Therefore, a gain or loss cannot be attributed solely to the attention operation: it may also arise from the inductive bias of dense recurrence. This is still a substantially stronger parameter-budget control than the 2,030-parameter bypass alone.

Implementation steps

1. Reuse Stage 6 data loading, optimization, multi-SNR schedule, checkpoint selection and evaluation code.
2. Extend the Stage 6 trainer to accept explicit `-recurrent-type` (`diagonal` or `dense`) and `-recurrent-hidden` (integer) arguments, passing them into the existing `Config` constructor **before** model instantiation. Keep `-attention-mode` (`full` or `bypass`). The original Stage 6 script may require a small modification to expose these options; do not assume the flags already work.
3. Confirm that bypass is implemented with a constant all-one mask or direct identity feature path, and that the classifier receives the same 20-channel encoded features as in the full model.
4. Print trainable parameter count before training and assert 4,322 for Arm C. Include `attention_mode`, `recurrent_type`, `recurrent_hidden` and validation metrics in checkpoint metadata.
5. Independently train Arm C from scratch with seeds 42, 123 and 2026. Reuse existing Stage 6 A and B checkpoints **only if** training/evaluation procedures and noise generation match exactly.
6. Keep λ = 0, epochs = 60, batch size = 8, segment length = 1.5 s, SNR sets −10, −5, 0, +5, +10 and +15 dB, and mean validation HTER checkpoint selection.
7. Compare paired seed-wise differences (C − A), mean and sample SD, per-SNR HTER, MR, FAR, and spike activity; preserve all raw checkpoint paths/results.
8. Use validation for architecture analysis. If performing a final test, decide configuration and seed aggregation before inspecting results. Earlier test exploration means claims of untouched test-set confirmation should be avoided.

Suggested commands once the new flags are implemented

```
for seed in 42 123 2026; do
  python stage6_train.py \
    --manifest real_data/speech_commands_pilot/manifest.jsonl \
    --attention-mode bypass \
    --recurrent-type dense \
    --recurrent-hidden 45 \
    --epochs 60 \
    --batch-size 8 \
    --segment-s 1.5 \
    --snrs -10 -5 0 5 10 15 \
    --val-snrs -10 -5 0 5 10 15 \
    --seed "$seed" \
```

```
    --output "runs/stage7_matched_bypass_seed${seed}.pt"
done
```

Warning: These commands describe the intended interface, not a verified executable extension of your local `stage6_train.py`. The new flags must be implemented first. No new training outputs have been generated by this plan.

Decision table template

Seed	Full attention HTER (%)	Matched bypass HTER (%)	Matched − full (pp)
42	14.731	pending	pending
123	14.393	pending	pending
2026	14.864	pending	pending
Mean	14.662	pending	pending

Interpretation

- If full attention consistently beats the equally sized no-attention model, it supports the view that its architecture is useful beyond raw parameter count in this implementation.
- If matched bypass matches or beats full attention, much of the Stage 6 benefit may be attributable to representational capacity or to architectural alternatives to attention.
- Neither outcome settles the paper's hardware power-efficiency claim; parameter count is not a hardware power measurement.

▼ Stage 8 Findings

1. Objective

Characterize the final performance of the models trained in Stages 6–7 across six SNR levels and three random seeds, and place those findings in the context of the original sVAD paper without equating unlike datasets or noise distributions. No new architecture was trained during Stage 8.

2. Evaluation design

- **Data:** Speech Commands pilot test split. Speech/non-speech intervals were estimated from the audio signal; they are pseudo-labels rather than manually verified boundaries.
- **Noise:** Synthetic additive Gaussian noise at −10, −5, 0, +5, +10, +15 dB SNR. The dataset recordings themselves may contain background noise.
- **Seeds:** 42, 123, 2026.
- **Metric:** Miss rate (MR), false alarm rate (FAR), and half total error rate, HTER = (MR + FAR)/2. Means are equally averaged over six SNR conditions and three seeds; standard deviation is calculated over per-seed six-SNR mean HTER values (sample SD).
- **Checkpoints:** Best validation-selected checkpoints from Stages 6–7; selected using average validation HTER across six SNRs.
- **Important evaluation status:** Earlier stages already inspected this test split. Consequently these results are characterization of an exposed test set, not unbiased untouched-holdout confirmation.

3. Compared architectures

Label	Description	Effective parameter budget
Full attention	Shared encoder, learned spiking attention, diagonal sRNN with 32 hidden neurons	4,313

Label	Description	Effective parameter budget
Light bypass	Shared encoder, no learned attention, diagonal sRNN with 32 hidden neurons	2,030
Matched bypass	Shared encoder, no learned attention, dense sRNN with 45 hidden neurons	4,322 expected; verify from checkpoint metadata

The matched comparison changes recurrence density and hidden width as well as attention, so it is a capacity-matched architecture comparison rather than an isolated causal test of attention.

4. Per-seed HTER results (%)

Architecture	Seed 42	Seed 123	Seed 2026	Three-seed mean	Sample SD (pp)
Full attention	14.125	13.390	14.230	**13.915**	**0.458**
Light bypass	14.133	13.428	15.387	14.316	0.992
Matched bypass	13.803	14.867	14.813	14.494	0.599

The user-provided summary from the evaluation script gives 14.495% and 0.598 pp for matched bypass. Small differences between these values and calculations from the displayed four-decimal per-SNR HTER values are rounding effects. For reporting the main summary, prefer the script's full-precision outputs: **full 13.915% (0.458 pp), light 14.316% (0.993 pp), matched 14.495% (0.598 pp)**.

Mean paired differences (bypass minus full; positive means full is better):

- Light bypass: +0.401 percentage points.
- Matched bypass: +0.580 percentage points.

These effects are descriptive; with only three paired seeds, they do not establish statistical significance.

5. Per-SNR mean HTER (%)

Means of the three reported per-seed values, rounded to two decimals.

SNR	Full attention	Light bypass	Matched bypass	Lowest HTER
−10 dB	20.06	20.84	**18.90**	Matched bypass
−5 dB	**16.12**	16.64	16.36	Full
0 dB	**13.35**	13.97	14.61	Full
+5 dB	**11.65**	12.03	12.81	Full
+10 dB	**11.01**	11.05	12.06	Full
+15 dB	**11.29**	11.37	12.23	Full

Per-SNR advantage of full attention versus light bypass: +0.78, +0.51, +0.62, +0.37, +0.04, +0.08 pp (ordered −10 through +15 dB; positive means full is better).

Per-SNR advantage of full attention versus matched bypass: **−1.16**, +0.23, +1.26, +1.16, +1.05, +0.94 pp. The negative value at −10 dB means the matched bypass is better under the most severe noise tested.

6. Error-rate behavior

Across the full-attention seeds, miss rate generally decreases with increasing SNR: at −10 dB, MR values are 0.3579, 0.3476 and 0.3730; at +15 dB they are 0.1554, 0.1360 and 0.1560. Meanwhile FAR at +15 dB rises to 0.0737, 0.0810 and 0.0752, versus 0.0495, 0.0358 and 0.0400 at −10 dB. Thus the HTER curve can turn upward at high SNR despite lower miss rates. This behavior merits discussion but cannot be attributed to a specific internal mechanism without additional measurements.

7. Interpretation

1. **Overall ranking:** Full attention gives the lowest three-seed average HTER among the tested models.

2. **Noise dependence:** The matched dense bypass has the strongest −10 dB average, so the attention benefit is not uniform across noise conditions.
3. **Capacity:** Full attention and matched bypass have nearly identical parameter budgets but different recurrent architectures. Their performance difference cannot uniquely isolate attention's contribution.
4. **Lightweight alternative:** The 2,030-parameter light bypass is competitive but less accurate on average; parameter count by itself does not establish energy use, speed or deployment efficiency.
5. **Reproducibility:** Full attention shows the lowest observed seed-to-seed SD in this sample; three seeds are too few for strong uncertainty statements.

8. Relationship to the original sVAD paper

Previously recorded figures for the paper's QUT-NOISE-TIMIT evaluation are low-noise 4.0%, medium-noise 11.9%, and high-noise 19.1% HTER. These values **must not** be presented as directly comparable to our numbers: the current work uses a Speech Commands pilot, generated Gaussian noise and estimated speech labels, rather than the authors' original dataset and environmental-noise protocol. Architectural details of the original implementation also remain uncertain. No Loihi hardware power measurement or direct hardware-energy validation has been performed.

9. Limitations and threats to validity

- Small pilot and three random seeds.
- Pseudo-ground-truth speech boundaries and potential label noise.
- Synthetic Gaussian noise instead of the paper's environmental noise distributions.
- Exposure of the test split earlier in the project; not an untouched holdout.
- Comparison confounded by changes to recurrent connectivity/width in capacity matching.
- SNR-dependent FAR behavior remains unexplained.
- No hardware power, latency or energy-per-inference measurements.
- Not a bit-for-bit reproduction of the published system.

10. Stage 8 conclusion and handoff

Stage 8 is complete as final descriptive evaluation. Across nine selected checkpoints, full attention reaches 13.915% average test-characterization HTER, compared with 14.316% for the lightweight bypass and 14.495% for the parameter-matched dense bypass. It wins on average and at five of six SNR levels, but the matched bypass wins at −10 dB. These small-sample findings favor full attention overall without proving a statistically significant benefit or reproducing the original paper's benchmark.

Stage 9: Consolidate Stages 1–8 into the final reconstruction report, with a traceable architecture specification, protocol, main results, reproducibility notes, unresolved implementation details, and a measured statement of what has and has not been reproduced. No further training is required for the current planned scope.